

DP-ES: Differentially Private Evolution Strategies for Prompt Optimization

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*A population-based DP prompt optimizer that spends privacy budget **only on scoring** — stable where token-level DP construction collapses.*

Outline: from a diagnosis to a structurally robust optimizer

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|----|--------------------|--|
| 01 | PROBLEM | Why prompt optimization on sensitive data must be private — and why token-level DP is unstable |
| 02 | IDEA | Decouple exploration from privacy spending |
| 03 | METHOD | Three phases per iteration and the formal update rule |
| 04 | RESULTS | Four tasks, stability under noise, ablations, a local model |
| 05 | PRIVACY | End-to-end guarantee, implementation checks, memorization stress test |
| 06 | EFFICIENCY & SCOPE | Cost profile, and what the evidence does not yet establish |
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Prompt optimization on sensitive data must be private — and stable

Automatic prompt optimization tunes prompts directly on task data — when those data are sensitive (patient records, financial logs), the optimizer can **memorize and expose** private information.

Formal **(ϵ , δ)-differential privacy** bounds this leakage — but the required noise stresses the optimizer itself: under a **tight budget** ($\epsilon \leq 1$), the leading token-level method, DP-OPT, becomes highly unstable.

THE HEADLINE NUMBER

49.5 $\pm$ 28.5%

DP-OPT accuracy on GSM8K, mean $\pm$ std over **30 runs** at ($\epsilon \leq 1.0$, $\delta = 10^{-5}$) — a ± 1 std band spanning roughly 21% to 78%.

HYPOTHETICAL

*What a **non-private** optimizer could produce: a final prompt that embeds memorized records — e.g. “...patient #12847, diagnosed with ..., prescribed ...” — shipped to every downstream user of the prompt.*

Setting: discrete prompts, black-box LLMs, record-level DP

OBJECTIVE

Private dataset $D = \{(x_i, y_i)\}_{i=1}^n$, prompt space P . Find

$$p^* = \arg \max_{p \in P} f(p, D)$$

PRIVACY REQUIREMENT

The optimizer M is (ϵ, δ) -DP if, for all neighbouring datasets $D \sim D'$ and all output sets S ,

$$\Pr[M(D) \in S] \leq e^\epsilon \cdot \Pr[M(D') \in S] + \delta$$

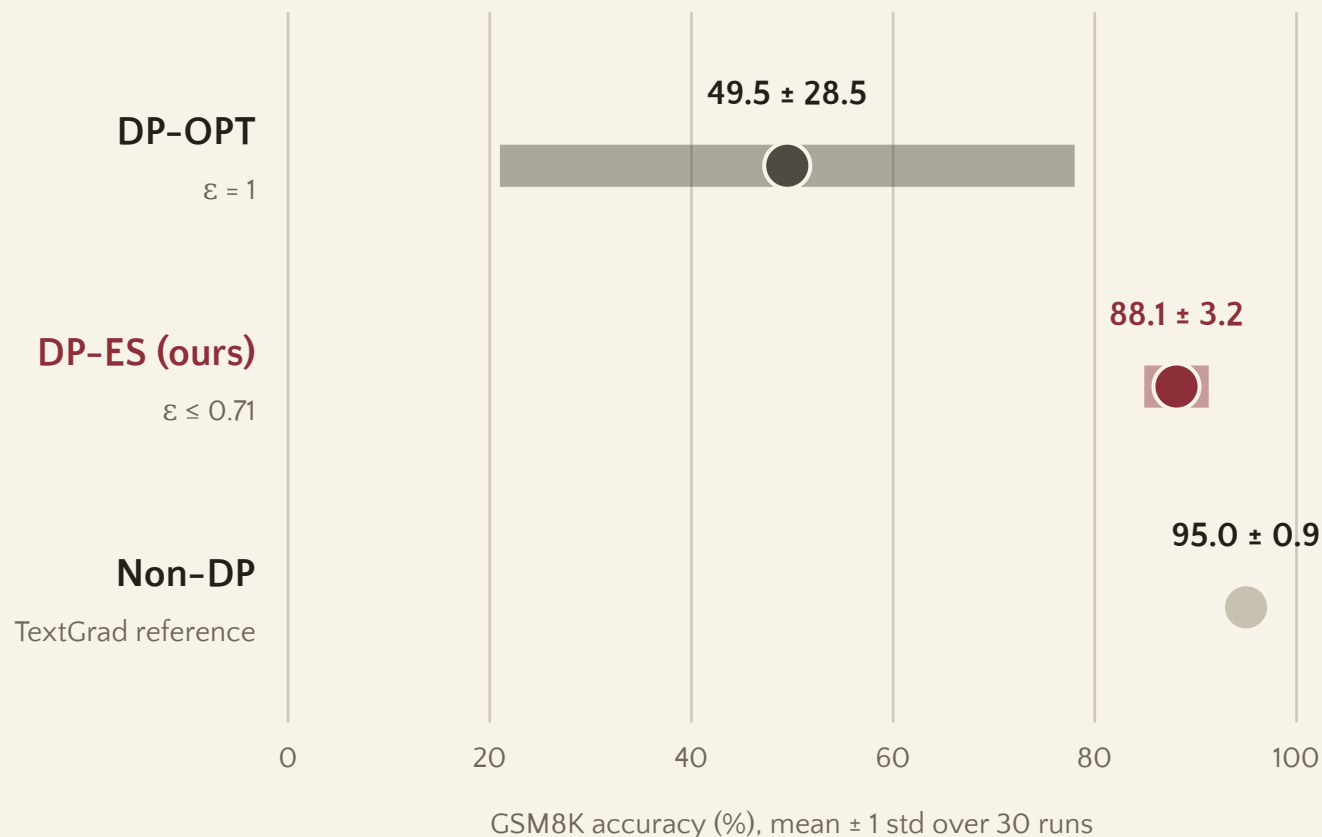
Privacy unit: one record (x_i, y_i) . Released object: the final prompt. The LLM is reached only through an API, so weight-based DP training (DP-SGD, DP-LoRA) does not apply.

WHERE DP-ES SITS

Method	Formal DP	Discrete	Mechanism
TextGrad	✗	✓	LLM “gradients”
OPRO	✗	✓	LLM scoring
EvoPrompt	✗	✓	Evolutionary search
PromptBreeder	✗	✓	Evolutionary search
DP-SGD	✓	✗	Gaussian noise
DP-OPT	✓	✓	Histogram + exp. mech.
DP-ES (ours)	✓	✓	Evolution + Gaussian

DP-ES adds formal record-level DP to evolutionary search over discrete prompts.

Token-level DP optimization is unstable under $\epsilon \leq 1$



RUN-TO-RUN SPREAD

± 28.5 pp

DP-OPT standard deviation on GSM8K: one run may succeed, the next may collapse.

WITH DP-ES

$\approx 9\times$ lower

standard deviation (3.2 pp) under a tighter recomputed bound.

DP-OPT: black-box adaptation of Hong et al. (ICLR 2024). LLM: DeepSeek-V3.2. We report full distributions, without a post-hoc failure threshold.

Diagnosis: why token-level DP construction collapses

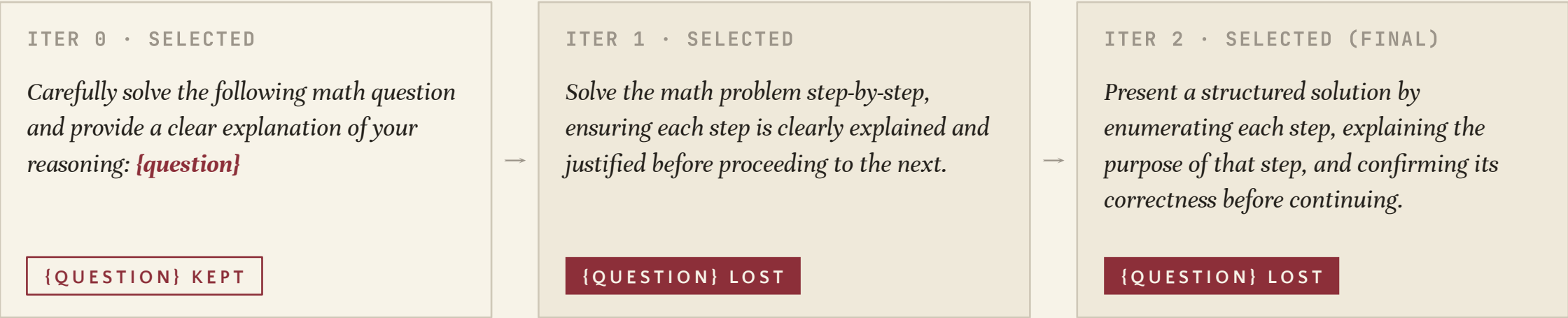
Greedy token-by-token construction samples each next token from privately aggregated histogram counts.

Noise-driven, irreversible choices. Privacy noise can flip a token choice; with no backtracking, one early decision redirects all later tokens.

Template drift. The constructed prompt silently loses the structural placeholder the task needs.

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candidate prompts omit the {question} placeholder in **one representative logged GSM8K trajectory** (2/3 selected prompts omit it) — evidence of **template drift**, not a failure rate.



Real DP-OPT log (seed 42 — a high-accuracy run, 91%). The evaluator appends the question when the placeholder is absent: a **structural-drift diagnostic**, not a zero-accuracy event.

Core idea: decouple exploration from privacy spending

Evolve a population of complete prompts.

*Pay privacy budget **only** for scoring.*

MUTATION privacy-free

An LLM rewrites complete prompt strings. It never accesses D — by the **post-processing property**, exploration can be arbitrarily rich at **zero privacy cost**.

SCORING touches D

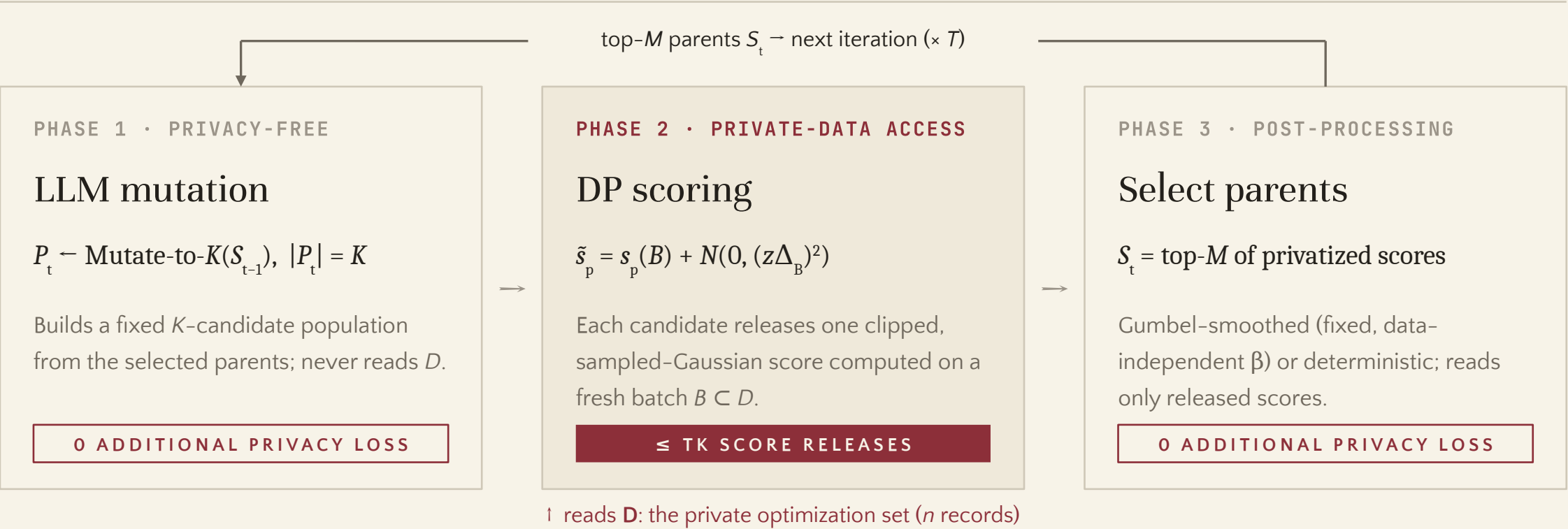
Sampled-Gaussian evaluation is the only step that reads the private dataset — the entire budget is concentrated here and composed by an **RDP accountant**.

SELECTION post-processing

Top- M parents chosen from **privatized scores** (deterministic or Gumbel-smoothed). Uses only released values — **zero additional privacy loss**.

This instantiates the Private Evolution principle from DP data synthesis (Lin et al., 2024; Xie et al., 2024) for prompt optimization.

DP-ES: three phases per iteration



Privacy accounting. Only Phase 2 accesses D . At most $T \cdot K$ sampled-Gaussian score releases compose in one RDP accountant; with $n = 200, b = 10, z = 10, K = 6, T = 3$, AutoDP gives $\epsilon \leq 0.71$ at $\delta = 10^{-5}$ — conservatively stated as $\epsilon \leq 1$. Mutation and selection add no privacy loss.

One iteration, formally

MUTATE
privacy-free

$P_t \leftarrow \text{Mutate-to-}K(S_{t-1}), \quad |P_t| = K$

The mutation LLM sees only parent prompts — never records or record-derived feedback.

SCORE
touches D

$u_i(p) = 1[\text{LLM}(p, x_i) = y_i], \quad s_p(B) = (1/b) \sum_{i \in B} \text{clip}(u_i(p), [0, C])$

$\tilde{s}_p = s_p(B) + N(0, \sigma^2), \quad \sigma = zC/b$

Fresh uniform batch $B \subset D, |B| = b$, per candidate; sensitivity C/b .

SELECT
post-processing

$\hat{s}_p = \tilde{s}_p + \text{Gumbel}(0, \beta)$ (default), or $\hat{s}_p = \tilde{s}_p$ (deterministic)

$S_t \leftarrow \text{top-}M(P_t, \hat{s});$ the best-so-far prompt p^* is tracked from $\tilde{s}$.

CONFIGURATION (ALL TASKS)

Symbol	Value	Meaning
K	6	candidates per round
M	2	parents kept
T	3	rounds (iterations)
b	10	scoring batch size
z	10	noise multiplier
C	1	clipping bound
n	200	private records
mode	—	BALANCED mutation

Fixed in advance; not selected using task utility.

Experimental setup: four tasks, one fixed configuration

GSM8K Multi-step math reasoning 200-example optimization set; full 1,319-example test set	MEDQA Medical multiple-choice QA 200-question USMLE subset; near-ceiling regime	BANKING77 Intent classification 77 classes, 200 examples; large label space	ALPACA Instruction following 200 instructions; LLM-judge score (DeepSeek-Chat)
MODELS DeepSeek-V3.2 via API (main results) Qwen2.5-7B-Instruct, local (validation)	BASELINES DP-OPT, black-box adaptation ($\epsilon = 1$) Non-DP TextGrad (a reference, not a bound) PromptDPSPGD, soft prompts (local)	PROTOCOL Common ceiling $\epsilon \leq 1$, $\delta = 10^{-5}$ 30 runs on GSM8K; 3 seeds elsewhere Same initial prompt for all methods	

Main results across four tasks ($\epsilon \leq 1.0$, $\delta = 10^{-5}$)

GSM8K · 30 RUNS

+38.6 pp

over DP-OPT, with $\approx 9\times$ lower standard deviation (3.2 vs 28.5)

99.7 ± 0.5

MedQA (+6.2 pp over DP-OPT), near the non-DP ceiling

Method	GSM8K	MedQA	BANK77	Alpaca†
Non-DP	95.0±0.9	100.0±0.0	75.0±2.8	91.7±0.6
DP-OPT	49.5±28.5	93.5±12.0	75.3±2.8	87.1±1.4
DP-ES	88.1±3.2	99.7±0.5	73.5±2.2	86.8±0.8
Δ (ES-OPT)	+38.6	+6.2	-1.8	-0.3

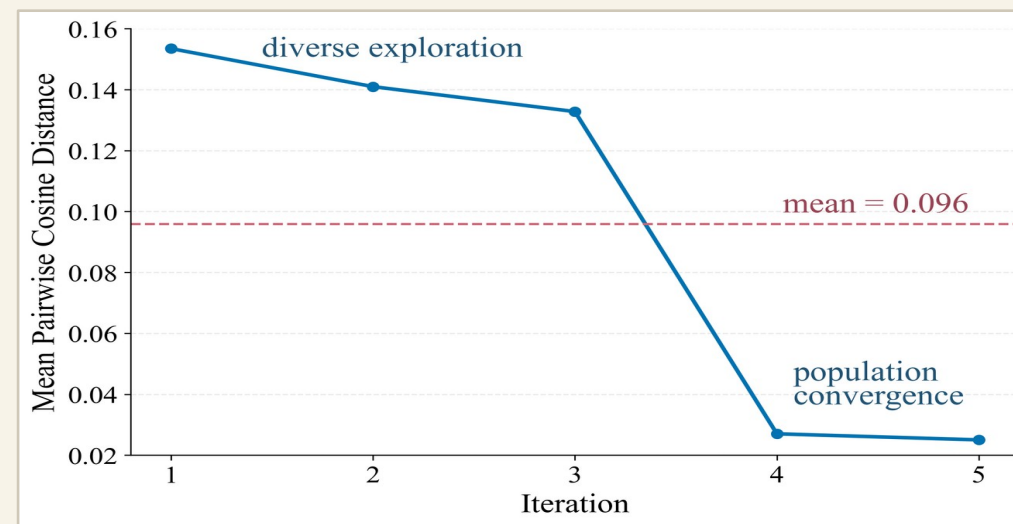
Mean $\pm$ std; 30 runs (GSM8K), three seeds (other tasks). All DP results under the stated ceiling $\epsilon \leq 1.0$; DP-ES's recomputed RDP bound is $\epsilon \leq 0.71$ at $\delta = 10^{-5}$. †Alpaca: LLM-judge ratings scaled to percentages. Largest gains where prompt structure matters; parity on BANKING77 and Alpaca.

Stability: population search survives noisy rankings

DP-ES does not eliminate noisy rankings — its population recovers from them.

Prompt-level mutations operate on **complete strings**, so the `{question}` placeholder survives the search (real log below).

Six candidates per round supply intact **alternatives** after noise-induced rank inversions.



Population diversity (mean pairwise prompt-embedding distance): high early, then converges.

FINAL DP-ES PROMPT · GSM8K · REAL LOG ($\epsilon \leq 1.0$)

`{QUESTION}` PRESERVED

*“You are a math tutor. Solve the question carefully and show reasoning: **{question}** (Provide detailed reasoning and justification.) (Be concise and accurate.)”*

Result: $88.1 \pm 3.2\%$ versus DP-OPT’s $49.5 \pm 28.5\%$ across 30 runs — complete alternatives persist across rounds.

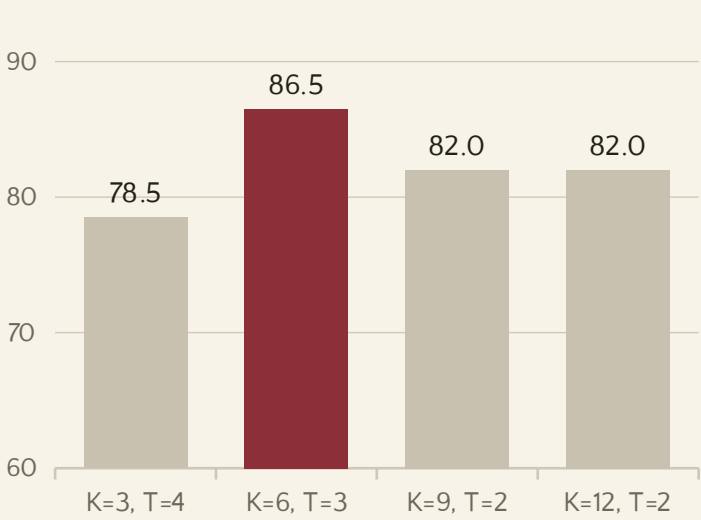
Ablations: selector, population size, and a local model

SELECTOR · GSM8K · 5 SEEDS

Selector	Accuracy (%)
Gumbel-smoothed	88.8 ± 3.4
Deterministic	88.6 ± 2.9

Both rules post-process privatized scores, so the privacy guarantee is identical; accuracy differs by only 0.2 pp. Gumbel smoothing stays the default, with no claimed variance advantage.

POPULATION SIZE · $K \cdot T \leq 24$



$K = 6, T = 3$ balances diversity and depth: small K explores too narrowly; large K leaves too few refinement rounds.

LOCAL QWEN2.5-7B · $\epsilon \leq 1$ · 3 SEEDS

Method	GSM8K	MedQA
DP-ES	53.8±6.1	81.2±1.3
DP-OPT	37.4±11.6	83.0±2.1
PromptDPSPGD	20.0±5.8	34.8±19.0
Non-DP	26.2±10.2	79.5±4.2

The reasoning gain transfers (+16.4 pp on GSM8K, half the std); MedQA is comparable – the main trend is not API-specific.

Formal guarantee: $\epsilon \leq 0.71$ at $\delta = 10^{-5}$

PROPOSITION 1

Mutation is privacy-free

Mutate sees only a public or previously privatized parent prompt, so by post-processing it consumes zero privacy budget.

PROPOSITION 2

Decoupled exploration

Privacy loss depends only on the score releases — not on the number or strength of mutations per iteration.

THEOREM

End-to-end (ϵ, δ) -DP

Composing $\leq TK$ sampled-Gaussian releases in Rényi DP (uniform subsampling without replacement) gives (ϵ, δ) -DP for the released prompt; selection is post-processing.

ACCOUNTING FOR THE REPORTED RUNS

$$n = 200, b = 10 \quad \rightarrow \quad q = b/n = 0.05$$

$$z = 10, C = 1 \quad \rightarrow \quad \sigma = zC/b = 1.0$$

$$K = 6, T = 3 \quad \rightarrow \quad \leq 18 \text{ releases}$$

AutoDP Rényi-DP accountant at $\delta = 10^{-5}$

$$\epsilon = 0.705$$

Reported conservatively as $\epsilon \leq 1$, the setting used for DP-OPT.

```
$ python scripts/compute_privacy.py
```

Implementation checks and a memorization stress test

Noise-distribution check. 1,000 injected-noise samples pass a Kolmogorov–Smirnov test against the configured Gaussian mechanism.

Stress-test setup. 200 synthetic customer profiles × 6 sensitive strings (name, phone, email, address, street fragment, account ID) = **1,200 strings**; we count exact and near-verbatim matches in the final prompt.

Adversarial control. The non-DP optimizer is shown raw records and told to embed a directory; DP-ES mutation never receives records.

NON-DP CONTROL

40

strings leaked (3.33%)

Final prompt (excerpt)

Use this customer directory:
Jessica Moore, (713) 907-8800,
jessica.moore83@mail.com;
Anthony Williams,
(312) 991-9095,
anthony.williams60@mail.com; ...

DP-ES

0

strings leaked (0.0%)

Final prompt (excerpt)

You are a customer service assistant. Retrieve relevant information based on user queries. If the query involves account information, verify before responding.

Synthetic profiles; excerpts and counts reproduced from the saved run.

Reading of the evidence. The guarantee comes from mechanism-level RDP accounting — attack-agnostic by construction. The stress test is a **targeted sanity check**: it is not membership inference and does not cover semantic leakage; finite audits cannot certify DP.

Efficiency: faster, with fewer private-data round trips

WALL-CLOCK (GSM8K, API)

2.5× faster

17.9 h vs 45.7 h for DP-OPT

PRIVATE-DATA ACCESS

3.3× fewer

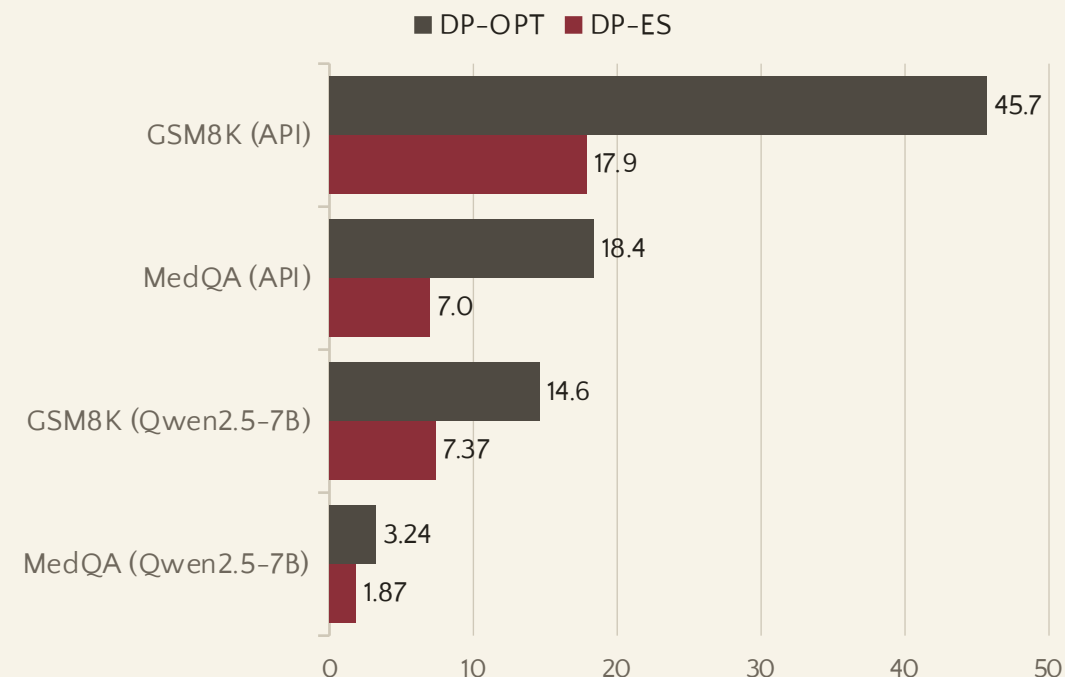
logged private-data call groups: 10 vs 33 (DP-ES mutations never touch D)

THE TRADE-OFF

2.8× more tokens

11,120 vs 4,016 in a representative run — DP-ES trades tokens for robustness

WALL-CLOCK HOURS · FOUR SETTINGS



Why faster? DP-OPT queries private aggregates while constructing tokens and while evaluating; DP-ES mutations never touch *D*. Fewer serial private-data round trips dominate latency, so wall-clock time drops even though DP-ES uses larger contexts — the advantage is not uniformly lower provider cost.

Scope: what we establish — and what remains open

SUPPORTED BY THE EVIDENCE

- + More robust optimization under DP noise than token-level construction, especially where prompt structure is critical
- + A formal record-level guarantee ($\epsilon \leq 0.71$, $\delta = 10^{-5}$) from mechanism-level RDP accounting
- + Lower wall-clock time and fewer private-data round trips than DP-OPT

OPEN — FUTURE WORK

- ? End-to-end validation on genuinely sensitive, non-saturated deployment data
- ? MedQA is near ceiling (Non-DP 100.0%), limiting its discriminative power
- ? Attacks: exact-match only — no membership inference or record reconstruction
- ? One privacy level ($\epsilon \leq 1$), not a densely sampled privacy-utility frontier
- ? Advantage is task-dependent: parity on BANKING77 and Alpaca

Next steps. Adaptive population sizing · multi-objective search over utility, privacy and stability · federated DP-ES for distributed sensitive data · a theory linking prompt search space to privacy budget.

TAKEAWAY

Decouple exploration from privacy: evolve complete prompts,
pay budget **only for scoring**.

$88.1 \pm 3.2\%$

GSM8K under $\varepsilon \leq 1.0$
30 runs · $\approx 9\times$ lower std

$\varepsilon \leq 0.71$

recomputed RDP bound
at $\delta = 10^{-5}$

$2.5\times$

faster wall-clock
than DP-OPT on GSM8K

0 strings

synthetic exact-match
stress test (200 profiles)

github.com/StephCpa/dp-es

Contact: liaiping@nudt.edu.cn

Happy to take questions.

Threat model and assumptions

TRUSTED CURATOR

The party running DP-ES may access D but releases only the final prompt.

RECORD-LEVEL PRIVACY

The unit is one record (x_i, y_i) ; group privacy scales ϵ with group size.

TRUSTED BOUNDARY

The evaluation endpoint is a trusted processor; DP protects the released prompt, not API inputs.

MUTATION ISOLATION

The mutation LLM sees only parent prompts. PII memorized in pretraining is outside this threat model.

POST-PROCESSING SAFETY

Downstream use of the released prompt adds no further loss about D .

AUDITS ARE NOT PROOFS

Empirical checks can expose implementation errors but cannot certify DP; the guarantee is mechanism-level.

Where the efficiency comes from

Cost component	DP-OPT	DP-ES	Ratio
<i>Logged private-data call groups</i>			
Mutation (token-level / privacy-free)	3 (DP)	0 (free)	—
Evaluation (Gaussian-noised)	30	10	3.0× fewer
Total private-data groups	33	10	3.3× fewer
<i>Token usage</i>			
Input tokens	1,309	6,120	4.7× more
Output tokens	2,707	5,000	1.8× more
Total tokens	4,016	11,120	2.8× more
<i>Wall-clock</i>			
Runtime (hours)	16.6	6.5	2.5× faster
Accuracy (single run)	91.0%	86.5%	—

GSM8K, seed 42, $\epsilon \leq 1$, 200-example optimization set. Counts are logger-level call groups, not individual requests.

READING THE TABLE

DP-OPT’s token construction queries private counts, so its mutation calls also enter the accountant.

DP-ES mutation is post-processing; only its evaluation groups touch records.

Fewer serial round trips dominate latency, so runtime drops even though DP-ES uses more tokens. Provider pricing and validation size can change the cost comparison.

Reproducing the privacy bound

```
$ python -m pip install -e .  
$ python scripts/compute_privacy.py  
epsilon=0.705168  
delta=1e-05  
releases=18  
sampling_rate=0.050000
```

Defaults: $n = 200$, $b = 10$, $z = 10$, $K = 6$, $T = 3$, $\delta = 10^{-5}$. Recompute whenever any of them changes: a larger KT raises both cost and composed privacy loss.

PRIVACY CONTRACT OF THE RELEASED CODE

- 01 Per-record utilities are clipped to a public interval.
- 02 Every data-dependent evaluation uses the sampled-Gaussian scorer; raw utilities are never released.
- 03 The number of releases is fixed in advance.
- 04 Mutation receives no private records or record-derived feedback.
- 05 Selection and all later computation use only privatized outputs.